GeoPops

Synthetic population generator that creates agents with attributes spatially connected in home, school, workplace, and group quarters (GQ) networks

Agents and Attributes

Home

School

Work

GQ

Starsim

Agent-based disease modeling software with customizable components:

ss.People()
ss.Network()
ss.Disease()
ss.Intervention()
ss.Analyzer()

Measles

Quarantine

Age & Vax Status Tracking

ss.People()



ss.Network()



ss.Disease()



ss.Intervention()



ss.Analyzer()



The **GeoPops** with **Starsim** framework enables detailed scenario modeling because for every agent, we know:

- Several demographic characteristics (age, gender, race/ethnicity)
- Where they live and where they go to school or work
- Who infected them, when, and where (home, school, work, GQ)
- When they transitioned disease compartments (e.g., S, I, R)

Think of the framework like a puzzle. Each puzzle piece is a different model component (e.g., people, networks, disease). By customizing the logic and parameters of each component, you determine the how the puzzle pieces fit together and what picture they make.

All data used to make a GeoPops population are **publicly available**. All GeoPops and Starsim code is **open source**. AI-coding assistants can be super helpful to explain how things work and how to implement your own customizations.

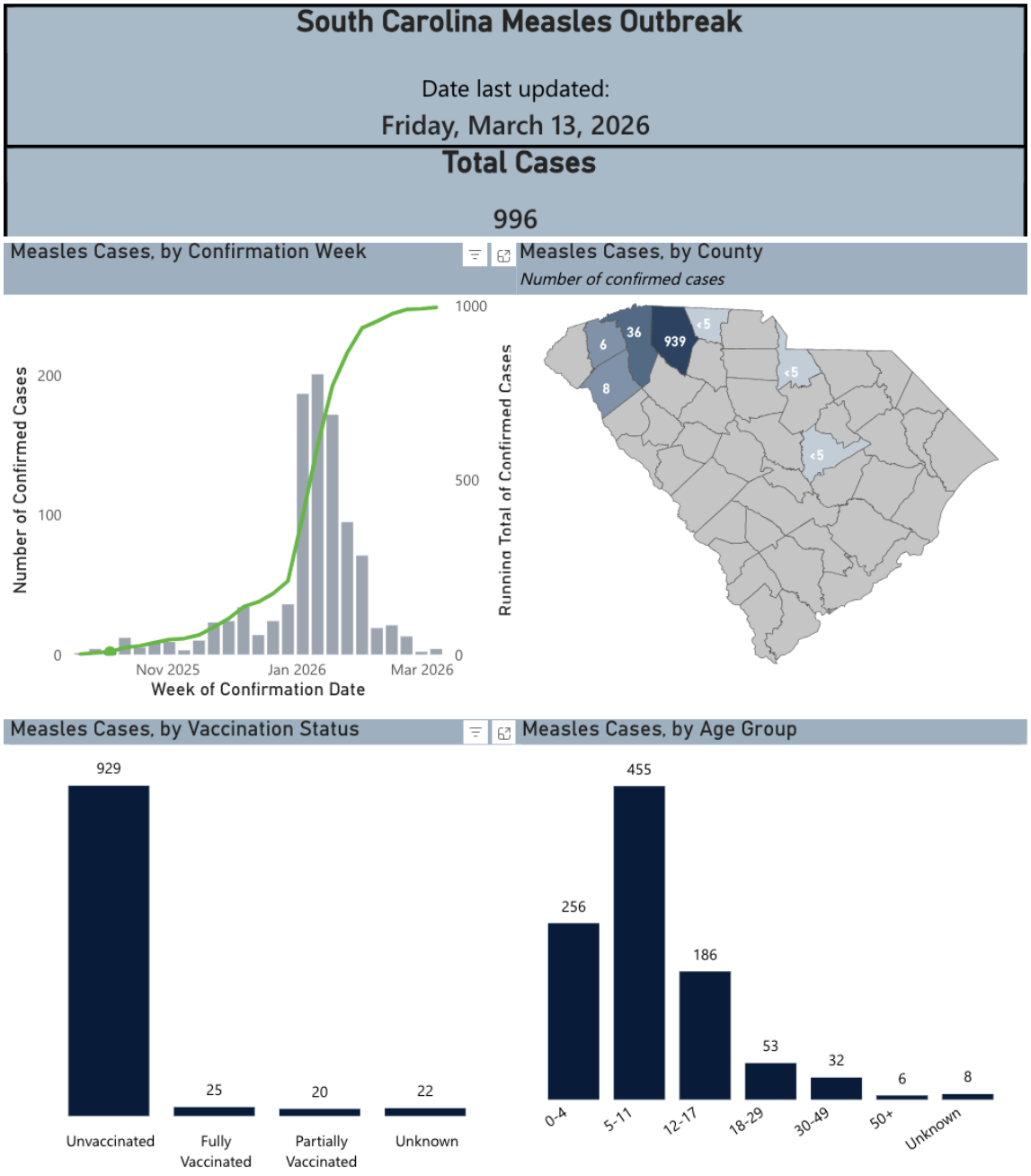
In this tutorial, we'll simulate a **measles outbreak** in Spartanburg County, SC. We'll demonstrate how to seed initial infections by school, simulate a quarantine strategy, and track infections by age and vaccination status. Don't be intimidated by the code! All the important stuff is in the text and figures!

1_run_geopops.ipynb	2_explore_people.ipynb	3_explore_networks.ipynb	4_measles_seeding.ipynb	5_measles_quarantine.ipynb

github.com/GeoPopsHub

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Screenshot from South Carolina Department of Public Health



On 2 Oct 2025, SC reported 8 measles cases from two schools in Spartanburg County. Since then, nearly 1,000 cases have been reported with the majority occurring in Spartanburg. *SC DPH Dashboard* (screenshot left)

In SC, 92.1% of kindergarten students had two doses of the MMR vaccine in 2023-2024, down from 95% in 2019-2020. *SC DPH*

Nationally, schools and healthcare providers are required to report measles laboratory confirmed cases to Departments of Public Health. *CDC*

"South Carolina doesn't require hospitals to report when they admit patients with measles-related illnesses. Available data show that only 2% of the state's measles cases have resulted in hospitalizations. Some infectious disease experts fear significant underreporting." *ProPublica*

The CDC describes the following illness progression. *CDC factsheet*, *CDC timeline*

Measles progression	Day	Model
Incubation, no symptoms	1-10	E
Incubation, cough/fever	11-14	I
Rash	15-18	I
Rash clears	19-21	R

Infected individuals are contagious around 4 days before and 4 days after developing a rash, which occurs around day 15. In our custom measles model, the exposed state (E) lasts 10 days on average, then agents are infectious (I) for 8 days on average.

Quarantine lasts 21 days after a known exposure, and there were 557 students quarantined as of 25 Jan 2026. *The State*, *SC DPH*